

RAHUL MITRA

SOFTWARE ENGINEER PROFILE

Singapore | mitrarahul2002@gmail.com | www.linkedin.com/in/rahulmitra-dev | github.com/Rah-Rah-Mitra | rahul-mitra.vercel.app

PROFILE SUMMARY

Early-career graduate software engineer with NUS ISE, Computer Science, and Mathematics training, building production-minded systems across Python, React, TypeScript, cloud deployment, AI-assisted development, and open-source tooling.

EDUCATION

National University of Singapore - Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics - Relevant coursework: Software Engineering; 3D Computer Vision; Artificial Intelligence; Data Structures and Algorithms; Simulation; Statistics; Operations Research	Aug 2023 - Jul 2027
Anderson Serangoon Junior College - Singapore-Cambridge GCE A-Level	Feb 2019 - Dec 2020

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd - Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.	Jun 2026 - Present
Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd - Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules. - Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.	Jan 2026 - Jun 2026
Software Engineer (Citizen Developer), People's Association - Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore. - Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks. - Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.	Sep 2025 - Present
Base Support Assistant, Republic of Singapore Navy - Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment. - Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.	Jan 2022 - Jan 2023

SELECTED PROJECTS AND COMPETITIONS

OnTheSpectrum - Codex and Blender MCP 3D Pipeline - Built an innovative local-first Blender-to-Three.js asset pipeline using Codex, Blender MCP, generated GLB assets, metadata, previews, playable world QA, and Remotion video creation workflows.	May 2026
AsyncDDGS - Open-Source Search Client - Maintained an asyncio-first, aiohttp-based DuckDuckGo search client on PyPI with pytest CI/CD and adoption across AI tooling, Discord bots, and search-enabled projects.	2024 - Present
Interactive Portfolio Infrastructure - Built a React, TypeScript, Three.js, Matter.js, and AI-assisted portfolio with profile switching, searchable field notes, analytics, and a 3D project world.	2026
SmartExam - GovTech x NTUPC Product Hackathon - Built a multi-agent autonomous exam generation system with RAG, Next.js, source ingestion, question generation, and education workflow automation.	2025

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Python; TypeScript; React; Next.js; Three.js; FastAPI; Flask; aiohttp; asyncio; Java; Docker; CI/CD; Azure; Singpass/Myinfo; Codex.
- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre - Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.	Sep 2025 - Present
Astronomy Head and Robotics Member, STEM Inc. (ASRJC) - Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.	2019 - 2020